

$$A = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 2 \\ 2 & 4 & 1 \end{bmatrix}$$

$$f_1 = a_1 = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

$$\|f_1\| = \sqrt{2^2 + 1^2 + 2^2} = 3$$

$$q_1 = \frac{f_1}{\|f_1\|} = \frac{1}{3} \cdot \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix}$$

$$c_2 \cdot q_1 = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix} = \begin{bmatrix} -17/9 \\ 14/9 \\ 10/9 \end{bmatrix} \xrightarrow{\times 9} f_2 = \begin{bmatrix} -17 \\ 14 \\ 10 \end{bmatrix}$$

$$\|f_2\| = \sqrt{\left(-\frac{17}{9}\right)^2 + \left(\frac{14}{9}\right)^2 + \left(\frac{10}{9}\right)^2} = \frac{\sqrt{65}}{3}$$

$$q_2 = \frac{f_2}{\|f_2\|} = \frac{3}{\sqrt{65}} \begin{bmatrix} -17/9 \\ 14/9 \\ 10/9 \end{bmatrix} = \begin{bmatrix} -17/3\sqrt{65} \\ 14/3\sqrt{65} \\ 10/3\sqrt{65} \end{bmatrix}$$

$$c_3 \cdot q_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix} = \frac{8}{3}$$

$$c_3 \cdot q_2 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \cdot \frac{1}{3\sqrt{65}} \begin{bmatrix} -17 \\ 14 \\ 10 \end{bmatrix} = \frac{1}{3\sqrt{65}}$$



$$f_3 = e_3 - (c_3 \cdot q_1) q_1 - (c_3 \cdot q_2) q_2 = \frac{11}{65} \begin{bmatrix} 2 \\ 6 \\ -5 \end{bmatrix}$$

$$\|f_3\| = \frac{11}{65} \cdot \sqrt{2^2 + 6^2 + (-5)^2} = \frac{11}{\sqrt{65}}$$

$$q_3 = \frac{f_3}{\|f_3\|} = \frac{1}{\sqrt{65}} \begin{bmatrix} 2 \\ 6 \\ -5 \end{bmatrix}$$

$$Q = \begin{bmatrix} \frac{2}{3} & \frac{-17}{3\sqrt{65}} & \frac{2}{\sqrt{65}} \\ \frac{1}{3} & \frac{14}{3\sqrt{65}} & \frac{6}{\sqrt{65}} \\ \frac{2}{3} & \frac{10}{3\sqrt{65}} & \frac{-5}{\sqrt{65}} \end{bmatrix}$$

$$R = \begin{bmatrix} 3 & \frac{13}{3} & \frac{8}{3} \\ 0 & \frac{\sqrt{65}}{3} & \frac{4}{3\sqrt{65}} \\ 0 & 0 & \frac{11}{\sqrt{65}} \end{bmatrix}$$